

```
Ubuntu 25.10 (Build_Complete_Pre_Install) [Running] - Oracle VirtualBox
File Machine View Input Devices Help

affan@affan-VirtualBox: ~/Desktop/CPU-Scheduling-Algorithms -- ./lab4
Mar 13 18:20

affan@affan-VirtualBox:~/Desktop$ git clone https://github.com/yousefkotp/CPU-Scheduling-Algorithms.git
Cloning into 'CPU-Scheduling-Algorithms'...
remote: Enumerating objects: 252, done.
remote: Counting objects: 100% (3/3), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 252 (delta 2), reused 1 (delta 1), pack-reused 249 (from 1)
Receiving objects: 100% (252/252), 396.13 KiB | 1021.00 KiB/s, done.
Resolving deltas: 100% (145/145), done.
affan@affan-VirtualBox:~/Desktop$ make
make: *** No targets specified and no makefile found. Stop.
affan@affan-VirtualBox:~/Desktop$ cd CPU-Scheduling-Algorithms/
affan@affan-VirtualBox:~/Desktop/CPU-Scheduling-Algorithms$ make
g++ -c main.cpp
g++ -o lab4 main.o
affan@affan-VirtualBox:~/Desktop/CPU-Scheduling-Algorithms$ ./lab4
trace
1
20
5
A,0,3
B,2,6
C,4,4
D,6,5
E,8,2
FCFS 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0
-----
A |*|*|*| | | | | | | | | | | | | | | | |
B | | | |*|*|*|*|*| | | | | | | | | | |
C | | | | | | | |*|*|*|*|*| | | | | | |
D | | | | | | | | | | | |*|*|*|*|*| | |
E | | | | | | | | | | | | | | | |*|*|
-----
affan@affan-VirtualBox:~/Desktop/CPU-Scheduling-Algorithms$ ./lab4
```

Test Case 1:

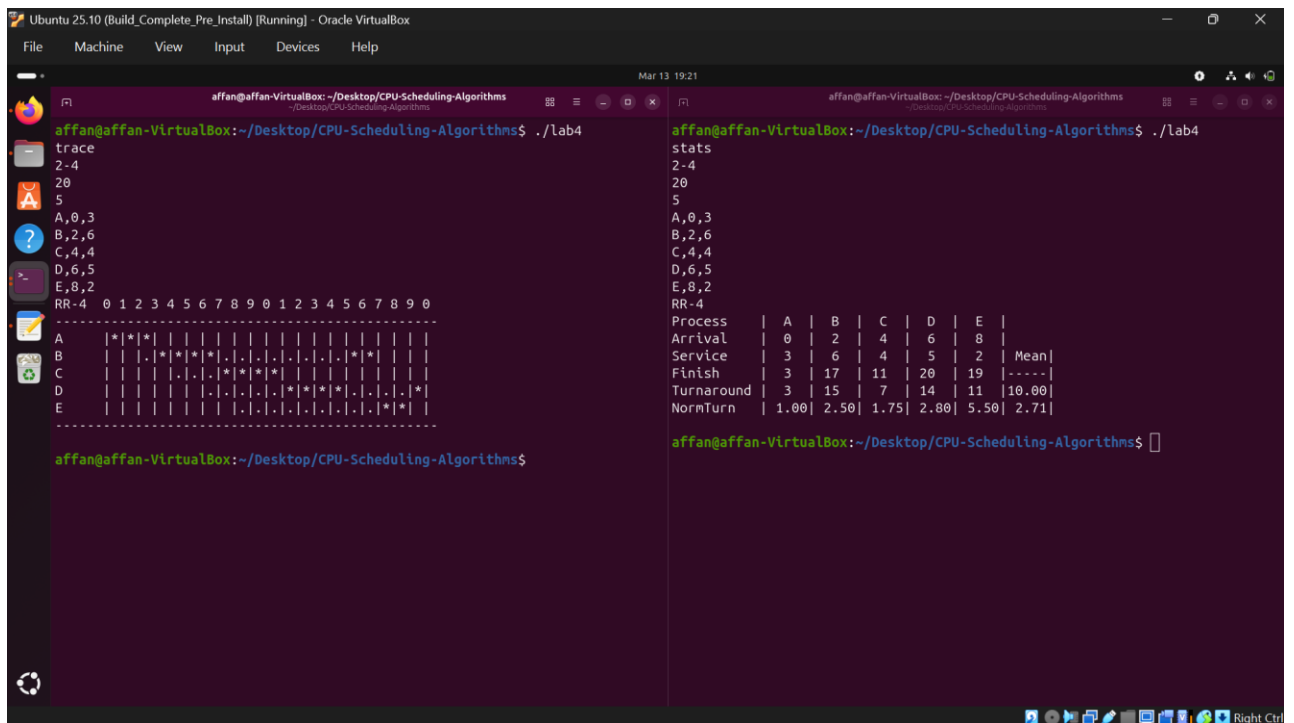
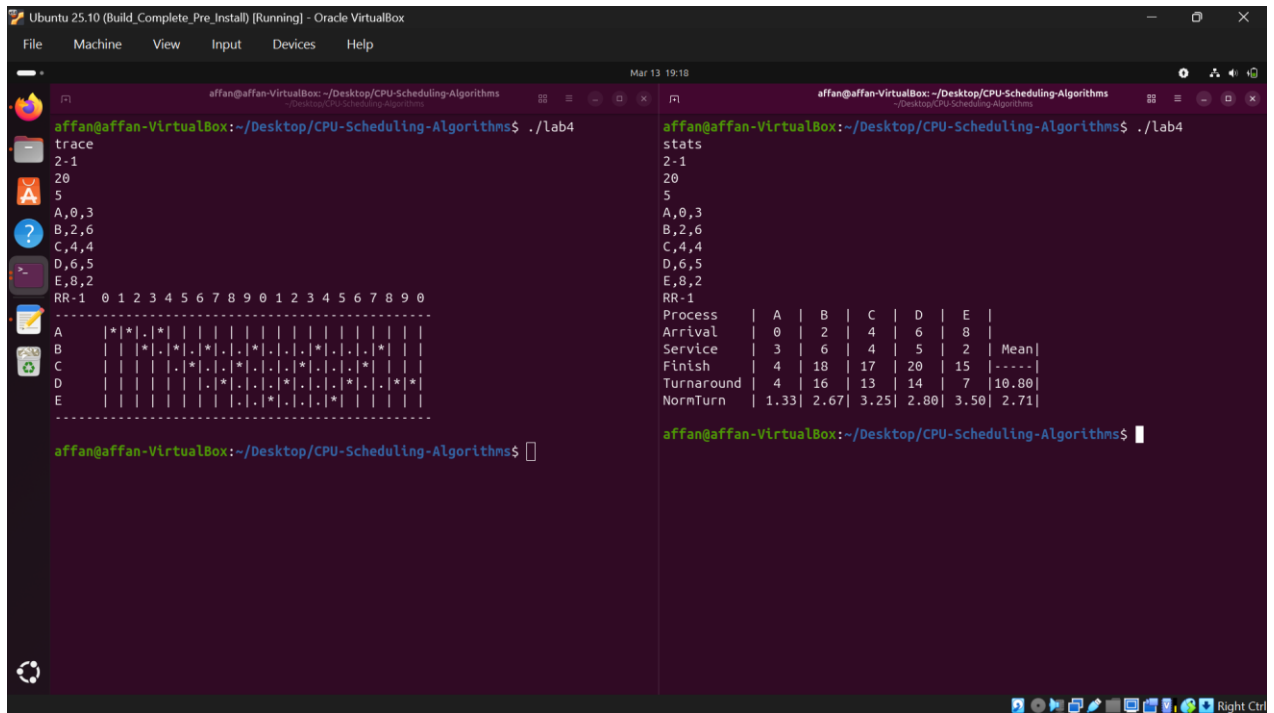
```
Ubuntu 25.10 (Build_Complete_Pre_Install) [Running] - Oracle VirtualBox
File Machine View Input Devices Help

affan@affan-VirtualBox: ~/Desktop/CPU-Scheduling-Algorithms
Mar 13 18:25

affan@affan-VirtualBox:~/Desktop/CPU-Scheduling-Algorithms$ ./lab4
stats
1
20
5
A,0,3
B,2,6
C,4,4
D,6,5
E,8,2
FCFS
Process | A | B | C | D | E |
Arrival | 0 | 2 | 4 | 6 | 8 |
Service | 3 | 6 | 4 | 5 | 2 | Mean |
Finish | 3 | 9 | 13 | 18 | 20 | ----- |
Turnaround | 3 | 7 | 9 | 12 | 12 | 8.60 |
NormTurn | 1.00 | 1.17 | 2.25 | 2.40 | 6.00 | 2.56 |

affan@affan-VirtualBox:~/Desktop/CPU-Scheduling-Algorithms$
```

|                 |
|-----------------|
| Test Cases 2,3: |
|-----------------|



24 K-0579

Date  
M T W T F S

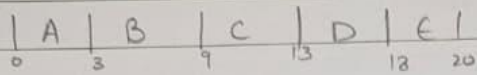
### TEST-CASE 1

| Processes | AT | BT | TAT | WT | RT |
|-----------|----|----|-----|----|----|
| A         | 0  | 3  | 3   | 0  | 0  |
| B         | 2  | 6  | 7   | 1  | 1  |
| C         | 4  | 4  | 9   | 5  | 5  |
| D         | 6  | 5  | 11  | 6  | 7  |
| E         | 8  | 2  | 12  | 10 | 10 |

$$\text{Avg TAT} = 8.8$$

$$\text{Avg WT} = 4.4$$

$$\text{Avg RT} = 4.6$$



### TC - 2

(TAT-DT)

| Processes | AT | BT | TAT | WT | RT |
|-----------|----|----|-----|----|----|
| A         | 0  | 3  | 4   | 1  | 0  |
| B         | 2  | 6  | 16  | 10 | 0  |
| C         | 4  | 4  | 13  | 9  | 1  |
| D         | 6  | 5  | 14  | 9  | 2  |
| E         | 8  | 2  | 7   | 5  | 2  |

Quantum = 1

$$\text{Avg TAT} = 10.8$$

$$\text{Avg WT} = 6.8$$

$$\text{Avg RT} = 1$$



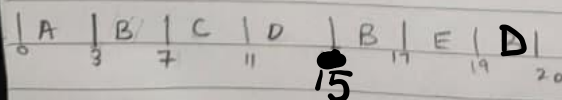
### TC-3

| Processes | AT | BT | TAT | WT | RT |
|-----------|----|----|-----|----|----|
| A         | 0  | 3  | 3   | 0  | 0  |
| B         | 2  | 6  | 15  | 9  | 1  |
| C         | 4  | 4  | 7   | 3  | 3  |
| D         | 6  | 5  | 14  | 9  | 5  |
| E         | 8  | 2  | 11  | 9  | 9  |

$$\text{Avg TAT} = 10$$

$$\text{Avg WT} = 4.8$$

$$\text{Avg RT} = 3.6$$



Test Cases 4,5,6:

The screenshot displays two terminal windows within an Oracle VM VirtualBox environment. The left window shows the initial state of a CPU scheduling algorithm, and the right window shows the final state with a Gantt chart and summary statistics.

**Left Terminal Window:**

```

affan@affan-VirtualBox:~/Desktop/CPU-Scheduling-Algorithms$ ./Lab4
trace
3
20
5
A:0,3
B:2,6
C:4,4
D:0,5
E:0,2
SPN  0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0
|-----|-----|-----|-----|-----|
B  | | | | | * | * | * | * | * | | | | | | | | | |
C  | | | | | | | | | | | | | * | * | * | * | * |
D  | | | | | | | | | | | | | | | | | | | | |
E  | | | | | | | | | | | | | | | | | | | | |
|-----|-----|-----|-----|-----|
affan@affan-VirtualBox:~/Desktop/CPU-Scheduling-Algorithms$

```

**Right Terminal Window:**

```

affan@affan-VirtualBox:~/Desktop/CPU-Scheduling-Algorithms$ ./Lab4
stats
3
20
5
A:0,3
B:2,6
C:4,4
D:0,5
E:0,2
SPN  0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0
|-----|-----|-----|-----|-----|
Process | A | B | C | D | E |
Arrival | 0 | 2 | 4 | 6 | 8 |
Service | 3 | 6 | 4 | 5 | 2 | Mean |
Finish  | 3 | 9 | 15 | 20 | 11 | ----|
Turnround | 3 | 7 | 11 | 14 | 3 | 1.60|
MinTurn | 1.00 | 1.17 | 2.75 | 2.80 | 1.50 | 1.84|
affan@affan-VirtualBox:~/Desktop/CPU-Scheduling-Algorithms$

```

The screenshot displays two terminal windows from a virtual machine named 'affan@affan-VirtualBox'.

**Left Terminal Window:**

```
affan@affan-VirtualBox: ~/Desktop/CPU-Scheduling-Algorithms
affan@affan-VirtualBox:~/Desktop/CPU-Scheduling-Algorithms$ ./lab4
Trace
4
20
5
A,0,3
B,2,6
C,4,4
D,6,5
E,8,2
SRT
0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0
-----
A | * | * |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
B |   |   | * |   |   |   |   |   |   |   | * | * | * | * |   |   |   |   |
C |   |   |   | * | * | * |   |   |   |   |   |   |   |   |   |   |   |
D |   |   |   |   |   |   | * |   |   |   |   |   |   |   |   | * | * |
E |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   | * |
-----
```

**Right Terminal Window:**

```
affan@affan-VirtualBox:~/Desktop/CPU-Scheduling-Algorithms$ ./lab4
stats
4
20
5
A,0,3
B,2,6
C,4,4
D,6,5
E,8,2
SRT
Process | A | B | C | D | E |
Arrival | 0 | 2 | 4 | 6 | 8 |
Service | 3 | 6 | 4 | 5 | 2 | Mean |
Finish | 3 | 15 | 8 | 20 | 10 | ----- |
Turnaround | 3 | 13 | 4 | 14 | 2 | 7.20 |
NormTurn | 1.00 | 2.17 | 1.00 | 2.80 | 1.00 | 1.59 |

affan@affan-VirtualBox:~/Desktop/CPU-Scheduling-Algorithms$
```

The image displays two terminal windows within an Oracle VM VirtualBox environment, running Ubuntu 25.10. Both windows are in the directory `~/Desktop/CPU-Scheduling-Algorithms`.

**Left Terminal Window:** The prompt is `affan@affan-VirtualBox:~/Desktop/CPU-Scheduling-Algorithms$ ./lab4`. The output shows a Gantt chart for a priority scheduling algorithm. Processes A, B, C, D, and E are scheduled based on their priority values (A:0, B:2, C:4, D:6, E:8). The chart shows the execution order and completion times for each process.

**Right Terminal Window:** The prompt is `affan@affan-VirtualBox:~/Desktop/CPU-Scheduling-Algorithms$ ./lab4`. The output shows a table of process metrics for the same set of processes. The metrics include Arrival, Service, Finish, Turnaround, and Normalized Turnaround times.

| Process    | A    | B    | C    | D    | E    |         |
|------------|------|------|------|------|------|---------|
| Arrival    | 0    | 2    | 4    | 6    | 8    |         |
| Service    | 3    | 6    | 4    | 5    | 2    | Mean    |
| Finish     | 3    | 9    | 13   | 20   | 15   | -----   |
| Turnaround | 3    | 7    | 9    | 14   | 7    | [ 8.00] |
| NormTurn   | 1.00 | 1.17 | 2.25 | 2.80 | 3.50 | 2.14    |

TC-4

Date 20  
M T W T F S S

| Processes | AT | BT | TAT | WT | RT |
|-----------|----|----|-----|----|----|
| A         | 0  | 3  | 3   | 0  | 0  |
| B         | 2  | 6  | 7   | 1  | 1  |
| C         | 4  | 4  | 11  | 7  | 7  |
| D         | 6  | 5  | 14  | 9  | 9  |
| E         | 8  | 2  | 3   | 1  | 1  |

| A | B | E | C | D |  
 0 3 9 11 15 20

Avg TAT = 7.6

Avg WT = 3.6

Avg RT = 3.6

TC-5

(Wrong output given)

| Processes | AT | BT | TAT | WT | RT |
|-----------|----|----|-----|----|----|
| A         | 0  | 3  | 3   | 0  | 0  |
| B         | 2  | 6  | 13  | 7  | 1  |
| C         | 4  | 4  | 4   | 0  | 0  |
| D         | 6  | 5  | 14  | 9  | 9  |
| E         | 8  | 2  | 2   | 0  | 0  |

Avg TAT = 7.2

Avg WT = 3.2

Avg RT = 2

| A | B | C | E | B | D |  
 0 3 4 9 10 15 20

TC-6

| Processes | AT | BT | TAT | WT | RT |
|-----------|----|----|-----|----|----|
| A         | 0  | 3  | 3   | 0  | 0  |
| B         | 2  | 6  | 7   | 1  | 1  |
| C         | 4  | 4  | 9   | 5  | 5  |
| D         | 6  | 5  | 14  | 9  | 9  |
| E         | 8  | 2  | 7   | 5  | 5  |

Avg TAT = 8

Avg WT = 4

Avg RT = 4

A | B | C | E | D |  
 0 2 9 13 15 20

## Test Case 7,8,9:

```

affan@affan-VirtualBox: ~/Desktop/CPU-Scheduling-Algorithms
affan@affan-VirtualBox:~/Desktop/CPU-Scheduling-Algorithms$ ./lab4
6
20
5
A,0,3
B,2,6
C,4,4
D,6,5
E,8,2
FB-1 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0
-----
A |*|.|.|*| | | | | | | | | | | | | | | | | |
B | | |*|.|.|*| | | | | | | | | | | | | | | |
C | | | | |*|.|.|*| | | | | | | | | | | | | |
D | | | | | | |*|.|.|*| | | | | | | | | | | |
E | | | | | | | | |*|.|.|*| | | | | | | | | |
-----

affan@affan-VirtualBox:~/Desktop/CPU-Scheduling-Algorithms$

```

```

affan@affan-VirtualBox:~/Desktop/CPU-Scheduling-Algorithms$ ./lab4
stats
6
20
5
A,0,3
B,2,6
C,4,4
D,6,5
E,8,2
FB-1
Process | A | B | C | D | E |
Arrival | 0 | 2 | 4 | 6 | 8 |
Service | 3 | 6 | 4 | 5 | 2 | Mean|
Finish | 4 | 20 | 16 | 19 | 11 | ----|
Turnaround | 4 | 18 | 12 | 13 | 3 | 10.00|
NormTurn | 1.33| 3.00| 3.00| 2.60| 1.50| 2.29|

affan@affan-VirtualBox:~/Desktop/CPU-Scheduling-Algorithms$

```

```

affan@affan-VirtualBox:~/Desktop/CPU-Scheduling-Algorithms
affan@affan-VirtualBox:~/Desktop/CPU-Scheduling-Algorithms$ ./lab4
7
20
5
A,0,3
B,2,6
C,4,4
D,6,5
E,8,2
FB-2l 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0
-----
A |*|.|.|*| | | | | | | | | | | | | | | | | |
B | | |*|.|.|*| | | | | | | | | | | | | | | |
C | | | | |*|.|.|*| | | | | | | | | | | | | |
D | | | | | | |*|.|.|*| | | | | | | | | | | |
E | | | | | | | | |*|.|.|*| | | | | | | | | |
-----

affan@affan-VirtualBox:~/Desktop/CPU-Scheduling-Algorithms$

```

```

affan@affan-VirtualBox:~/Desktop/CPU-Scheduling-Algorithms$ ./lab4
stats
7
20
5
A,0,3
B,2,6
C,4,4
D,6,5
E,8,2
FB-2l
Process | A | B | C | D | E |
Arrival | 0 | 2 | 4 | 6 | 8 |
Service | 3 | 6 | 4 | 5 | 2 | Mean|
Finish | 4 | 17 | 18 | 20 | 14 | ----|
Turnaround | 4 | 15 | 14 | 14 | 6 | 10.00|
NormTurn | 1.33| 2.50| 3.50| 2.80| 3.00| 2.63|

affan@affan-VirtualBox:~/Desktop/CPU-Scheduling-Algorithms$

```

```

affan@affan-VirtualBox:~/Desktop/CPU-Scheduling-Algorithms
affan@affan-VirtualBox:~/Desktop/CPU-Scheduling-Algorithms$ ./lab4
2-2
20
5
A,0,3
B,2,6
C,4,4
D,6,5
E,8,2
RR-2 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0
-----
A |*|.|.|*| | | | | | | | | | | | | | | | | |
B | | |*|.|.|*| | | | | | | | | | | | | | | |
C | | | | |*|.|.|*| | | | | | | | | | | | | |
D | | | | | | |*|.|.|*| | | | | | | | | | | |
E | | | | | | | | |*|.|.|*| | | | | | | | | |
-----

affan@affan-VirtualBox:~/Desktop/CPU-Scheduling-Algorithms$

```

```

affan@affan-VirtualBox:~/Desktop/CPU-Scheduling-Algorithms$ ./lab4
stats
2-2
20
5
A,0,3
B,2,6
C,4,4
D,6,5
E,8,2
RR-2
Process | A | B | C | D | E |
Arrival | 0 | 2 | 4 | 6 | 8 |
Service | 3 | 6 | 4 | 5 | 2 | Mean|
Finish | 5 | 17 | 13 | 20 | 15 | ----|
Turnaround | 5 | 15 | 9 | 14 | 7 | 10.00|
NormTurn | 1.67| 2.50| 2.25| 2.80| 3.50| 2.54|

affan@affan-VirtualBox:~/Desktop/CPU-Scheduling-Algorithms$

```



Date \_\_\_\_\_ 20\_\_\_\_  
M T W T F S S

TC - 7

| Processes | AT | BT | TAT | WT | RT |
|-----------|----|----|-----|----|----|
| A         | 0  | 3  | 4   | 1  | 0  |
| B         | 2  | 6  | 18  | 12 | 0  |
| C         | 4  | 4  | 14  | 10 | 0  |
| D         | 6  | 5  | 13  | 8  | 0  |
| E         | 8  | 2  | 4   | 2  | 0  |

$$\text{Avg TAT} = 10.6$$

$$\text{Avg WT} = 6.6$$

$$\text{Avg RT} = 0$$

TC - 4

| Processes | AT |
|-----------|----|
| A         | 0  |
| B         | 2  |
| C         | 4  |
| D         | 6  |
| E         | 8  |

A | B |  
0 3 4

TC - 8

| Processes | AT | BT | TAT | WT | RT |
|-----------|----|----|-----|----|----|
| A         | 0  | 3  | 4   | 1  | 0  |
| B         | 2  | 6  | 15  | 9  | 0  |
| C         | 4  | 4  | 11  | 7  | 0  |
| D         | 6  | 5  | 14  | 9  | 0  |
| E         | 8  | 2  | 6   | 4  | 0  |

$$\text{Avg TAT} = 10$$

$$\text{Avg WT} = 6$$

$$\text{Avg RT} = 0$$

TC - 5

| Processes | AT |
|-----------|----|
| A         | 0  |
| B         | 2  |
| C         | 4  |
| D         | 6  |
| E         | 8  |

A | B | C | E  
0 3 4 8

TC - 9

| Processes | AT | BT | TAT | WT | RT |
|-----------|----|----|-----|----|----|
| A         | 0  | 3  | 5   | 2  | 0  |
| B         | 2  | 6  | 15  | 9  | 0  |
| C         | 4  | 4  | 9   | 8  | 1  |
| D         | 6  | 5  | 14  | 9  | 3  |
| E         | 8  | 2  | 15  | 8  | 6  |

$$\text{Avg TAT} = 10.5$$

$$\text{Avg WT} = 6$$

$$\text{Avg RT} = 1.5$$

A | B | A | C | B | D | C | E | B | D |  
0 2 4 5 7 9 11 13 15 17 20

TC - 6

| Processes | AT |
|-----------|----|
| A         | 0  |
| B         | 2  |
| C         | 4  |
| D         | 6  |
| E         | 8  |

A | B | C | E  
0 2 4 8

## Test case 11,12

```
Ubuntu 25.10 (Build_Complete_Pre_Install) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Mar 13 19:46
affan@affan-VirtualBox: ~/Desktop/CPU-Scheduling-Algorithms
affan@affan-VirtualBox:~/Desktop/CPU-Scheduling-Algorithms$ ./lab4
trace
8-1
20
5
A,0,3
B,2,6
C,4,4
D,6,5
E,8,2
Aging 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0
-----
A  |*|*| | |*| | |*| | |*| | |*| |
B  | |*|*|*| |*| |*| | |*| | |*| |
C  | | | |*| | | | |*| | |*| | |*| |
D  | | | | | | |*| | |*| | | |*| |
E  | | | | | | | | | |*| | | |*| |
-----

affan@affan-VirtualBox:~/Desktop/CPU-Scheduling-Algorithms$
```

```
Ubuntu 25.10 (Build_Complete_Pre_Install) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Mar 13 19:47
affan@affan-VirtualBox: ~/Desktop/CPU-Scheduling-Algorithms
affan@affan-VirtualBox:~/Desktop/CPU-Scheduling-Algorithms$ ./lab4
trace
8-1
20
2
A,0,2
B,2,4
Aging 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0
-----
A  |*|*| | |*| | |*| | |*| | |*| |
B  | |*|*|*|*|*|*|*|*|*|*|*|*|
-----

affan@affan-VirtualBox:~/Desktop/CPU-Scheduling-Algorithms$
```



e 20  
W T F S

TC-10 → All previous outputs

Date 20  
M T W T F S S

TC-11, TC-12 'concerns aging scheduling, not  
mentioned in manual.

8  
4  
4